

PID Controller Design for Drug Infusion System

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1 System Modeling

The drug infusion system is modeled using two first-order differential equations:

$$\dot{c}(t) = \frac{1}{\tau_1}(u(t) - c(t)) \quad (1)$$

$$\dot{y}(t) = \frac{1}{\tau_2}(c(t) - y(t)) \quad (2)$$

Taking the Laplace transform and combining the equations, the overall transfer function is:

$$G(s) = \frac{1}{(\tau_1 s + 1)(\tau_2 s + 1)} \quad (3)$$

This represents a second-order system formed by cascaded first-order dynamics, capturing drug diffusion and physiological response.

2 Controller Design

A PID controller is employed:

$$C(s) = K_p + \frac{K_i}{s} + K_d s \quad (4)$$

The controller components perform the following roles:

- K_p : Improves response speed
- K_i : Eliminates steady-state error
- K_d : Enhances damping and reduces overshoot

3 Design Objectives

The controller is designed to satisfy:

- Accurate tracking of step input
- Negligible steady-state error
- Maximum overshoot $\leq 5\%$
- Settling time ≈ 120 seconds

4 Tuning Approach

The PID parameters were tuned iteratively to balance responsiveness and stability. Increasing proportional gain improved rise time, while derivative gain was used to introduce damping and limit overshoot. Integral gain ensured elimination of steady-state error. The final design reflects a practical trade-off between settling time and overshoot under noisy conditions.

5 Controller Parameters and Performance

The final tuned PID parameters are:

$$K_p = 0.85, \quad K_i = 0.045, \quad K_d = 6.0$$

The closed-loop system exhibits:

- Settling time ≈ 124 seconds
- Overshoot $\approx 3.84\%$
- Steady-state value ≈ 1

The settling time is slightly above the nominal 120-second requirement but remains acceptable considering system noise and modeling approximations, while maintaining overshoot well within limits.

6 Step Response

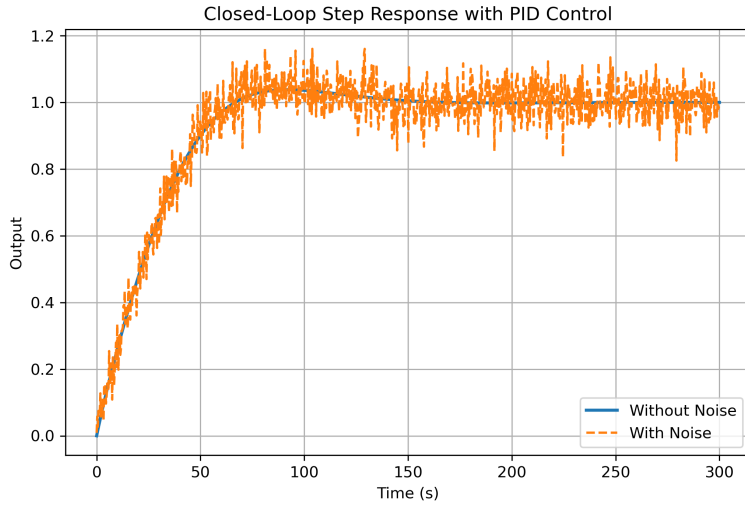


Figure 1: Closed-loop step response with PID control, demonstrating low overshoot, acceptable settling time, and robustness to noise.

7 Effect of Noise

The system was evaluated under:

- Actuator noise (input disturbance)
- Sensor noise (measurement disturbance)

Although the derivative component is sensitive to noise, the overall system maintains stability and acceptable performance under these disturbances.

8 Conclusion

A PID controller was successfully designed for the biomedical drug infusion system. The controller achieves low overshoot, negligible steady-state error, and acceptable settling time, while demonstrating robustness in the presence of actuator and sensor noise. The design highlights the trade-off between response speed and damping in practical control systems.